

WASP-1b

R-1288

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_171021 · created 2026-07-31T15:54:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458047.7477 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.015
Transit depth (Rp/Rs)^2	1.21 +/- 0.33 [%]
Orbital Inclination (inc)	86.61 +/- 1.39
Airmass coefficient 1 (a1)	1.007 +/- 0.01
Airmass coefficient 2 (a2)	-0.007 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.04 %
Best Comparison Star	#1 - [408, 121]
Optimal Aperture	4.52
Optimal Annulus	13.81
Transit Duration (day)	0.1472 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.11 ± 0.015 vs 0.1036 (deviation 0.26σ)
Duration fit vs published	0.1472 d vs 0.1567 d (deviation 0.85σ)
Mid-time O–C	0.7 min
Depth SNR	4.5
Residual scatter	1.04 %

Light curve

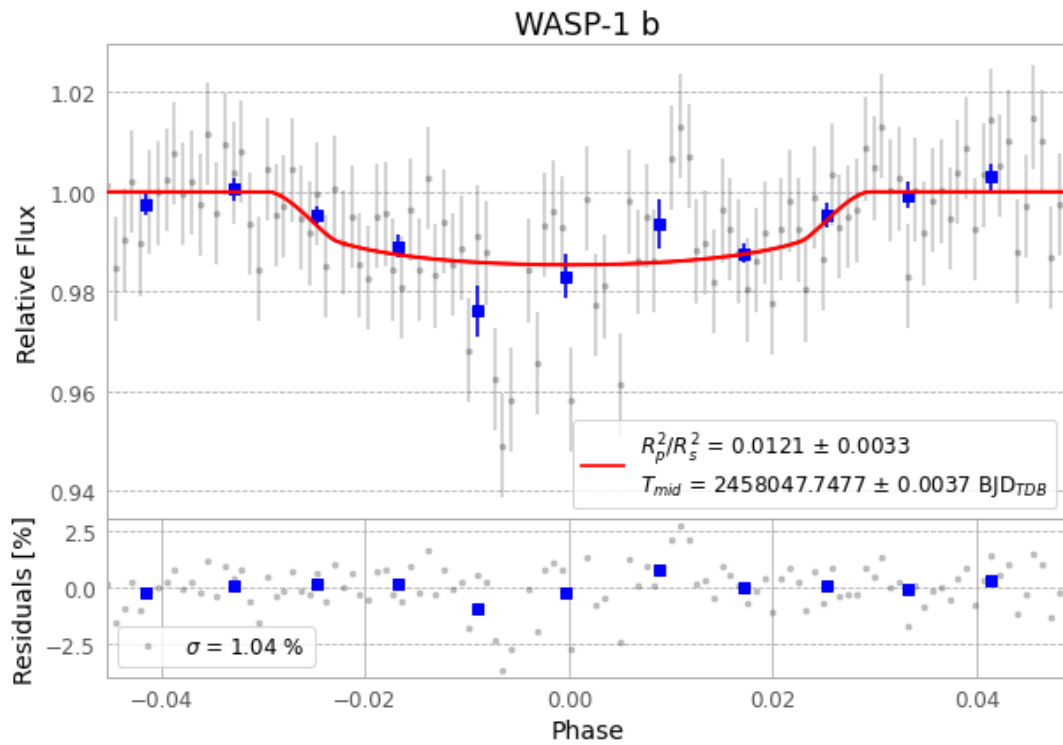


Figure 1 — FinalLightCurve_WASP-1 b_2017-10-20.png (SHA-256 5f82e83d07578ae7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b033ffb973df569b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

115 FITS frames (76 MB), WASP-1171021030312.FITS → WASP-1171021084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-171021.log (23ac1ca83566...), FinalLightCurve_WASP-1 b_2017-10-20.png (5f82e83d0757...), FinalLightCurve_WASP-1 b_2017-10-20.csv (251ed3fcec6...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 15:54Z.